THE LOCUS WHERE A PROPER FLAT MORPHISM IS AN ISOMORPHISM IS OPEN

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11814-11833, immediately following the statement at 11791-11812.

Editorial note. The author's parenthetical remark at the end of the proof is retained verbatim.

Editorial note. Two same-chapter lemmas and one cross-chapter lemma are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- The locus where a morphism is a closed immersion. chapter lemma lemma-where-closed-immersion, cited at proof line 11820. Supplies an open subspace W_2 of Z factoring exactly the base changes for which f becomes a closed immersion.
- The locus where a morphism is surjective and flat. chapter lemma lemma-where-surjective-flat, cited in the proof. Supplies an open subspace W_1 factoring exactly the base changes for which f becomes surjective flat.
- A surjective flat closed immersion of schemes is an isomorphism. Morphisms, lemma morphisms-lemma-characterize-flat-closed-immersions, cited at proof lines 11829-11830. Converts 'surjective flat closed immersion' into 'isomorphism' for the representing scheme morphism.

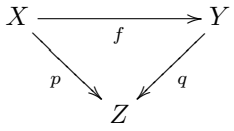
Source reference. Original lemma 05XA: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11642–11642. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05XC: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11729–11729. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04PW: focused-seeds/geometry/S9-spaces-where-isomorphism-spaces/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Consider a commutative diagram*



of algebraic spaces. Assume that

(1) *p is locally of finite presentation,*
(2) *p is flat,*
(3) *p is universally closed,*
(4) *q is locally of finite type,*
(5) *q is closed, and*
(6) *q is separated.*

Then there exists an open subspace $W \subset Z$ such that a morphism $Z' \to Z$ factors through W if and only if the base change $f_{Z'} : X_{Z'} \to Y_{Z'}$ is an isomorphism.

Proof. By Lemma 05XC there exists an open subspace $W_1 \subset Z$ such that $f_{Z'}$ is surjective and flat if and only if $Z' \to Z$ factors through W_1. By Lemma 05XA there exists an open subspace $W_2 \subset Z$ such that $f_{Z'}$ is a closed immersion if and only if $Z' \to Z$ factors through W_2. We claim that $W = W_1 \cap W_2$ works. Certainly, if $f_{Z'}$ is an isomorphism, then $Z' \to Z$ factors through W. Hence it suffices to show that f_W is an isomorphism. By construction f_W is a surjective flat closed immersion. In particular f_W is representable. Since a surjective flat closed immersion of schemes is an isomorphism (see Morphisms, Lemma 04PW) we win. (Note that actually f_W is locally of finite presentation, whence open, so you can avoid the use of this lemma if you like.) □